

Prometheus configuration — Helix Proxy (P7 observability)

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:** **config PLAN — NOT YET LIVE.** This directory defines the *intended* Prometheus scrape topology for the VPN-aware dynamic-routing extension. The scrape targets do **not** exist yet; they are built in later phases. Per Constitution §11.4.6 (no-guessing) nothing here claims metrics are flowing. **Authority:** Inherits the Helix Constitution submodule per §11.4.35. **Design source:** docs/superpowers/specs/2026-06-30-vpn-aware-proxy-extension-design.md (§11③ observability), docs/DYNAMIC_ROUTING.md, .env.example.

What this is

File	Purpose
prometheus.yml	Scrape config: prometheus self-scrape, helix-control-plane (/metrics on METRICS_PORT 34090), squid-exporter (boynux/squid-exporter on :9301).
squid-exporter.md	Which exporter image, what it scrapes (Squid mgr: cache-manager), and its port.

What is built when (honest phasing — §11.4.6)

- **Control-plane** /metrics (job helix-control-plane) → built in **P6** (control-API + admin UI). The planned custom metrics (helix_proxy_vpn_up, helix_proxy_tunnel_down_responses_total, ...) are emitted there.
- **squid-exporter** (job squid-exporter) → container wired in **P7/P10**.
- **Compose/integration wiring** (service names, network, Prometheus container itself) → owned by **P10**. The proxy-* target hostnames in prometheus.yml are forward references finalized there.

What IS provable now vs. what is owed to P10

- **Provable now (this phase):** the scrape config is **structurally valid** — promtool check config exits 0. Evidence: qa-results/p7-observability/promtool_check.txt.
- **NOT proven now (owed to P10):** that any target is reachable or scrapable, that any metric actually flows, or that the Grafana dashboard renders real data. promtool validates STRUCTURE ONLY — it never contacts a target. The live scrape + "dashboard renders real metrics" proof is a P10 deliverable (§11.4.6 / §11.4.123 — no metadata-only PASS substituted for runtime proof).

Re-validate locally (structural only, rootless §11.4.161)

```
podman run --rm -v "$PWD/config/prometheus:/c:ro,Z" \
  --entrypoint promtool prom/prometheus:latest \
  check config /c/prometheus.yml
# exit 0 == structurally valid (does NOT prove any target is up)
```